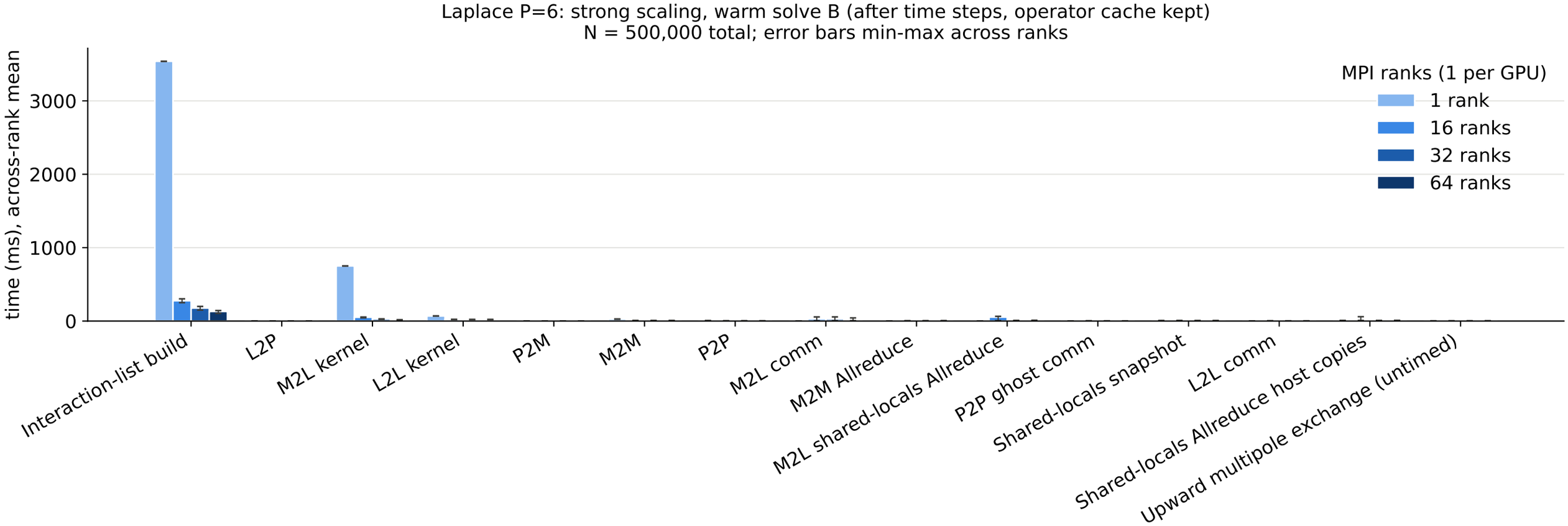
Laplace P=6: strong scaling, warm solve B (after time steps, operator cache kept)
N = 500,000 total; error bars min-max across ranks
time (ms), across-rank mean
0
1000
2000
3000
MPI ranks (1 per GPU)
1 rank
16 ranks
32 ranks
64 ranks
Interaction-list build
L2P
M2L kernel
L2L kernel
P2M
M2M
P2P
M2L comm
M2M Allreduce
M2L shared-locals Allreduce
P2P ghost comm
Shared-locals snapshot
L2L comm
Shared-locals Allreduce host copies
Upward multipole exchange (untimed)